Intro to Circuits

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CS 22 2021

May 21st, 2021

Overview

Basics

Addition

Expanded gates

Logic and circuits

Why AND/OR/NOT?

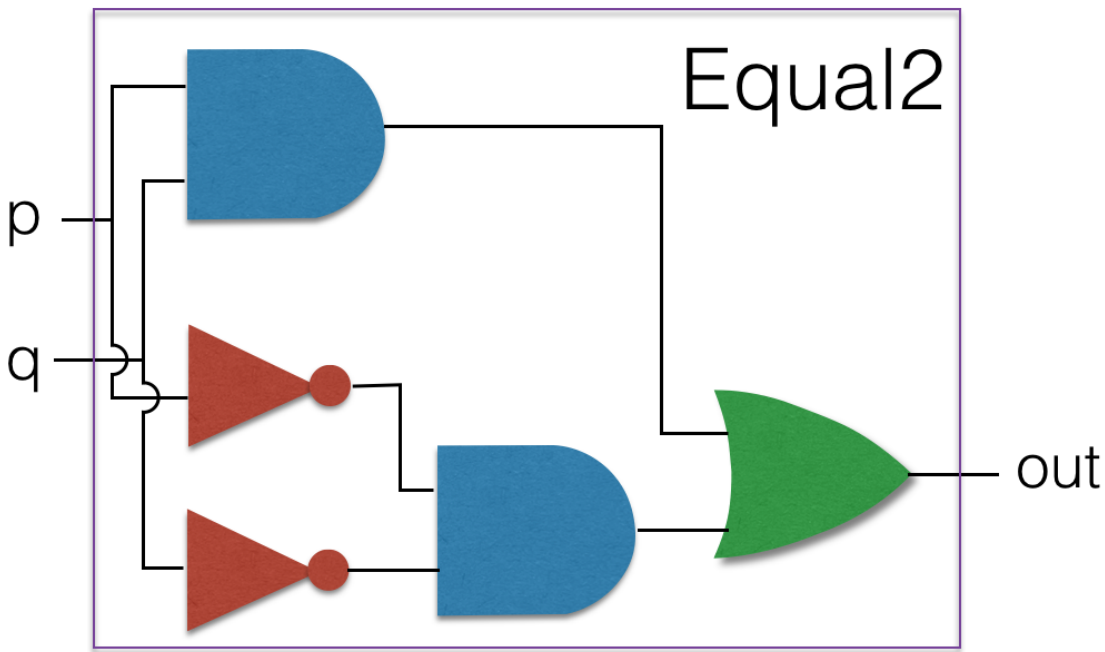
- Universal set.
- Already central to the study of logic.
- Can be implemented in physical hardware.

A *transistor* is roughly a voltage controlled switch and can be used to construct AND/OR/NOT. Voltage high: True. Voltage low: False.

To a first approximation, digital circuits are giant logic expressions. A computer is a giant logical expression whose outputs at one time step become the inputs at the next time step.

Basic Circuit Components

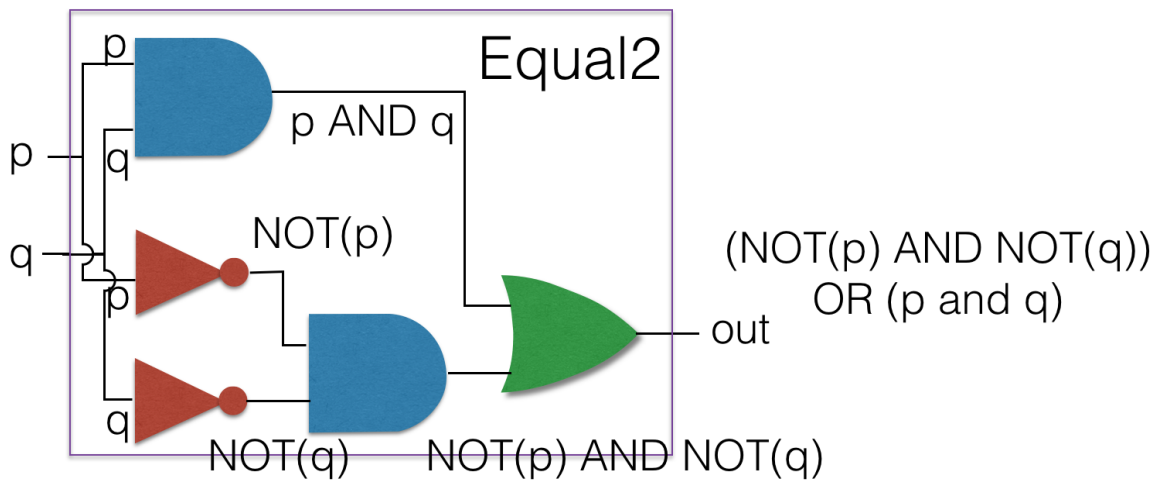
AND gates drawn as rectangles with a semicircular cap. OR gates similar, but concave rounded bottom and pointy top. NOT gates are triangles with a circle on the point.



How many AND/OR/NOTs?

Circuit Flow

Wires show where values propagate to. We can label each wire with an expression relating the inputs of the circuit to the value propagating on that wire.

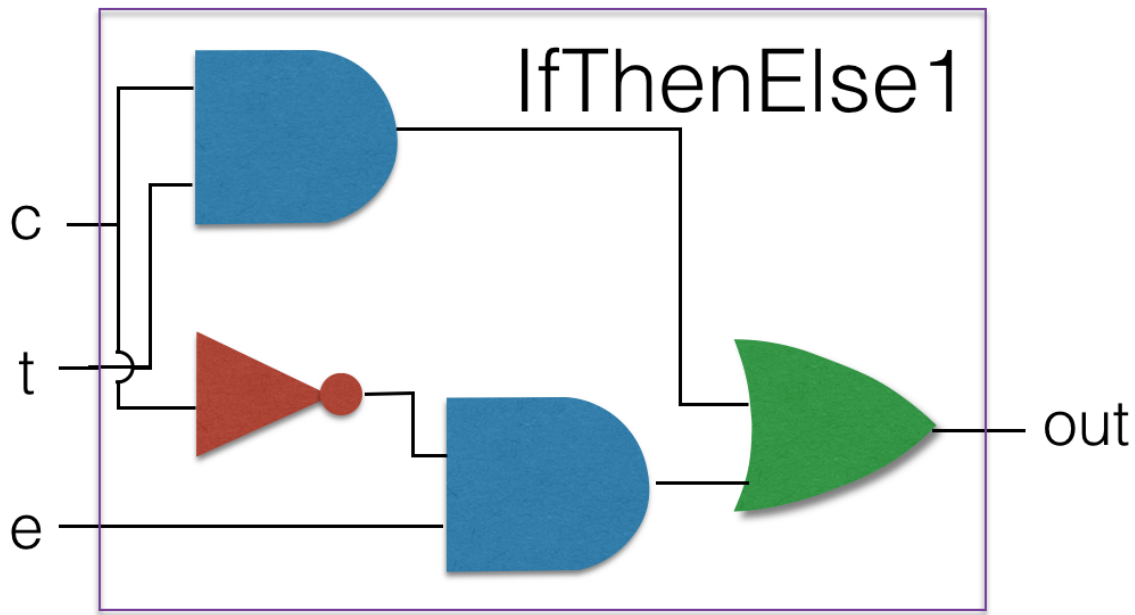


This circuit outputs (p AND q) OR (NOT(p) AND NOT(q)).

We'll call this circuit "equal2" since it is testing the equality of 2 truth values or two bits. Another name?

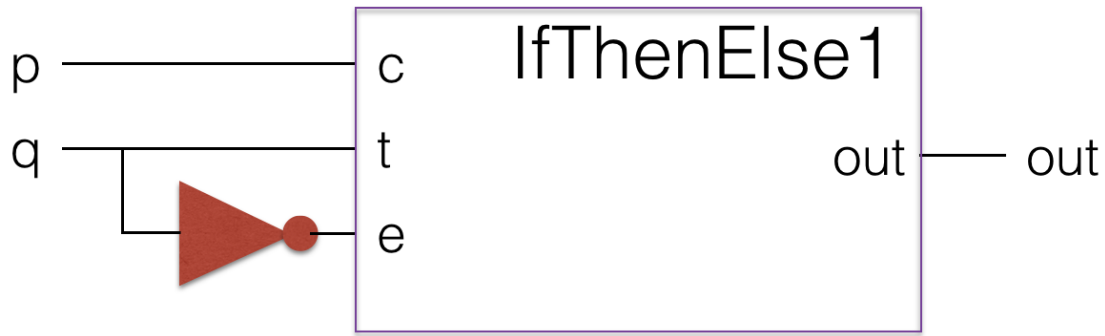
IfThenElse1

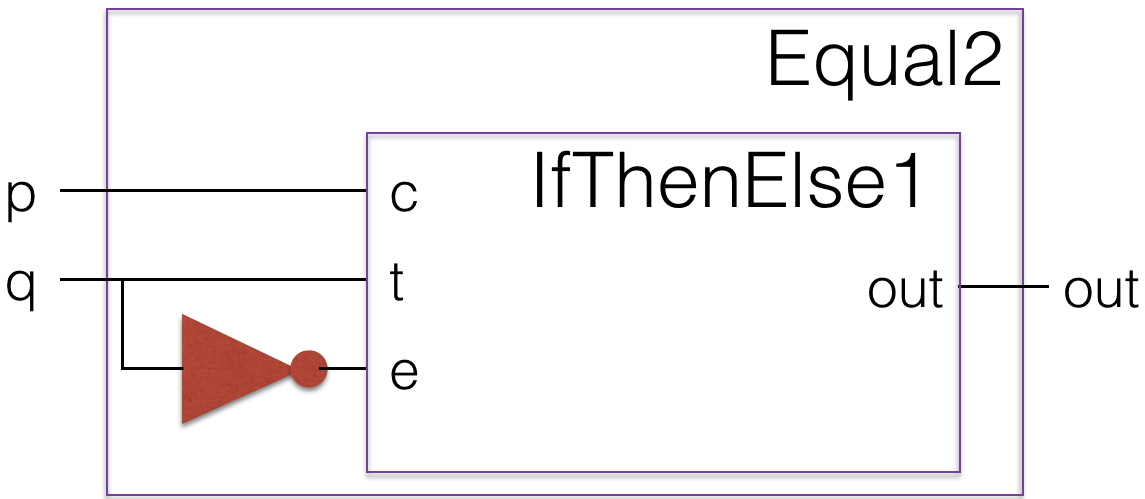
Here's another useful circuit. It outputs either input t or input e depending on the value of condition c.



Nested circuits

We can use IfThenElse1 to make another circuit. What is it?





Nested circuits 2

Here's another variant of IfThenElse1. What is it?

Truth and numbers

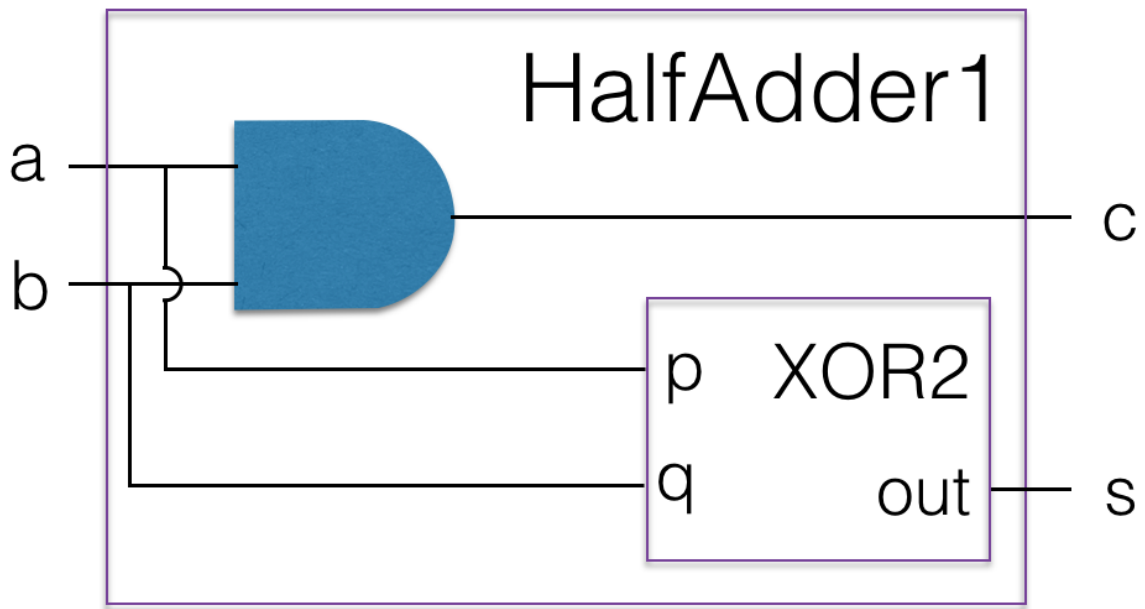
We can think of **T** as being 1 and **F** as being 0 and create circuits that compute on numbers. Let's add two bits.

		c	s
a	b	$a+b$	$a+b$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$c = a$ AND b. $s = a$ XOR b.

Half adder

Here's a circuit for adding two bits.



Adding two two-bit numbers

We can use the classic addition algorithm, modified to base 2.

Let's compute $11 + 01$ (aka $3 + 1$ in binary).

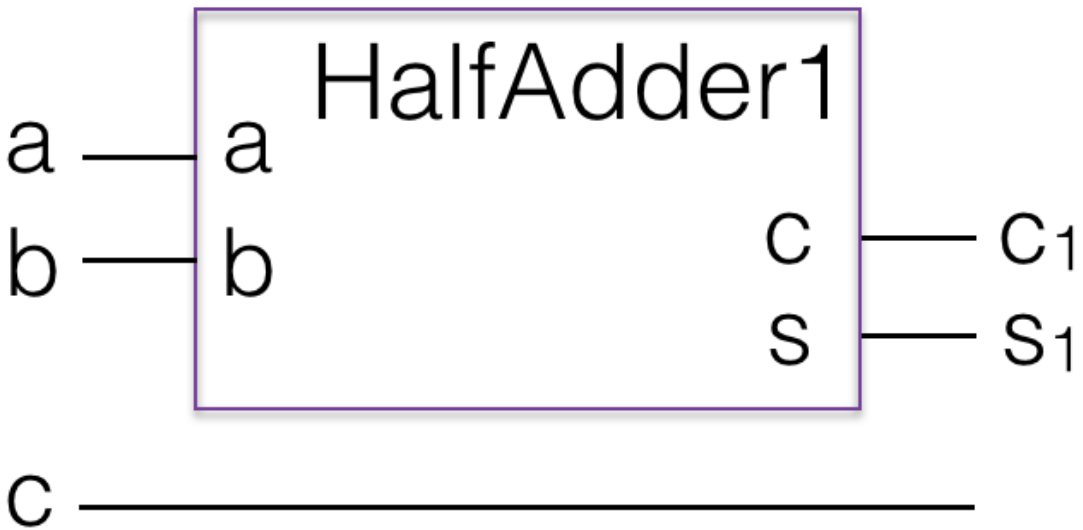
$$\begin{array}{cccc} & & 1 & 1 \\ + & & 0 & 1 \\ \hline \end{array}$$

$$\begin{array}{cccc} & & 1 & \\ & & 1 & 1 \\ + & & 0 & 1 \\ \hline & & & 0 \end{array}$$

$$\begin{array}{cccc} & & 1 & \\ & & 1 & 1 \\ + & & 0 & 1 \\ \hline 1 & 0 & 0 \end{array}$$

Full adder

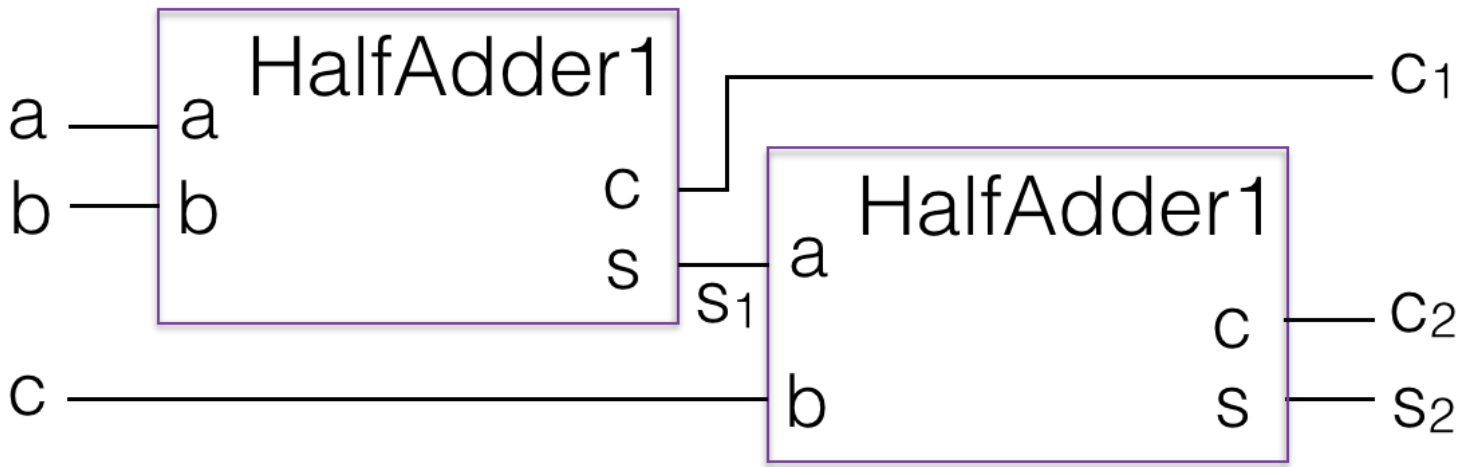
To handle this computation, we need to add *three* 1-bit numbers, not just two: $a + b + c$. Can add the first two bits to get a two-bit result.



Now, we need to add a two-bit number and a one bit number.

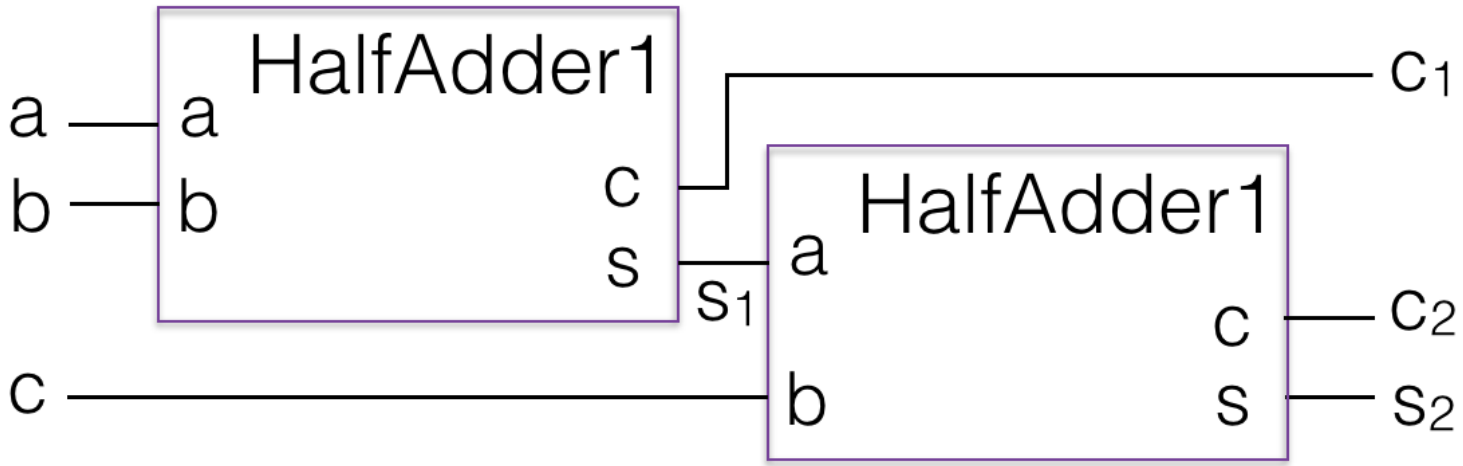
Partial full adder

We have $c_1 s_1 + c$. Not quite clear what to do. Could add the low order bit s_1 to c...



The low order bit of the sum s_2 is correct now. But, we have two carries, c_1 and c_2, each worth two...

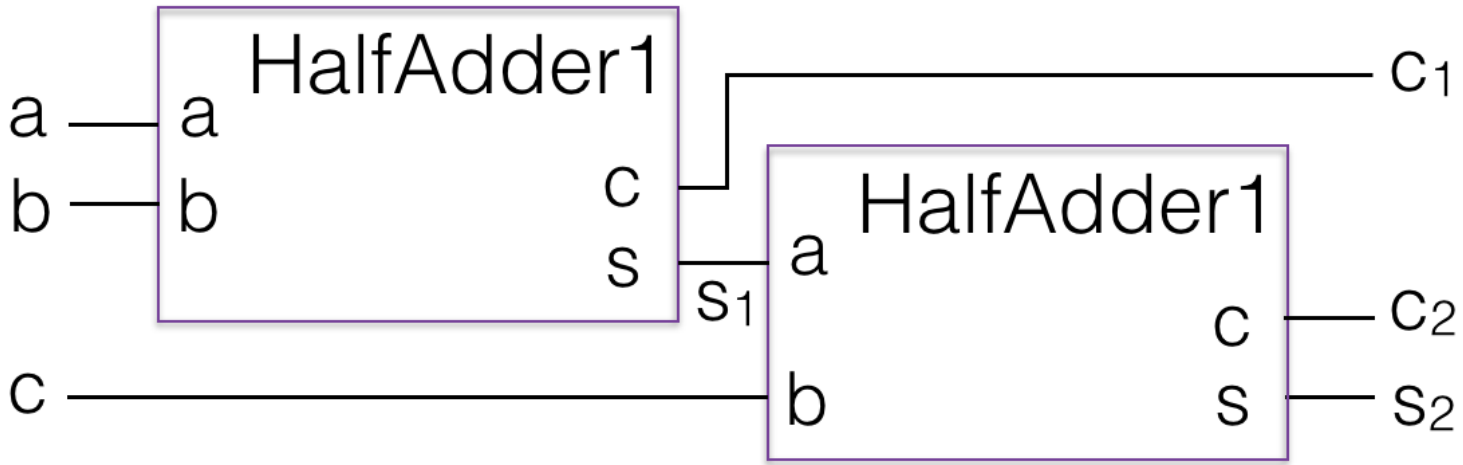
Towards the full full adder



Need to add the two carries c_1 and c_2. They are in the two's place. Could throw on another HalfAdder1. But:

- If a and b are both 1, their sum is 10. Thus, the carry c_1 is a one, but the sum s_1 is a zero. Adding in c won't cause a second carry in c_2.
- On the other hand, if at least one of a and b are less than 1, the bottom carry c_2 might be 1, but the top one c_2 is zero.

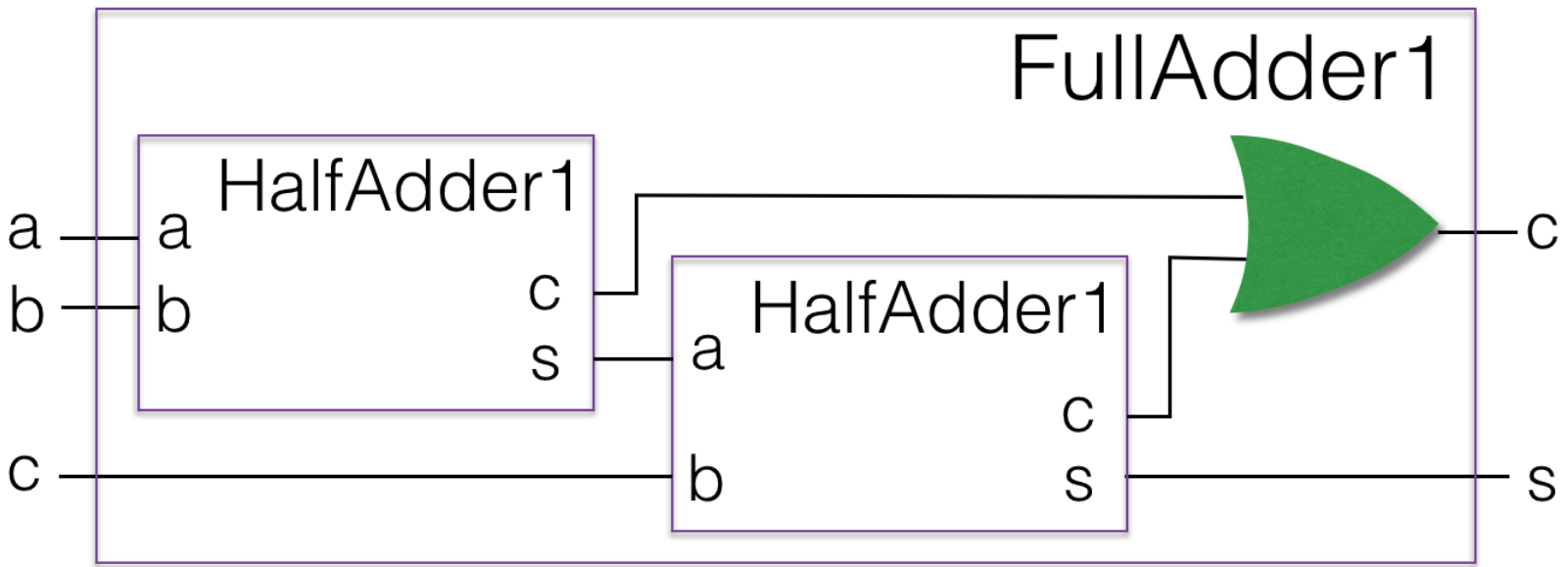
Truth table for adding the carries



c_1	c_2	$c_1 + c_2$
0	0	0
0	1	1
1	0	1
1	1	—

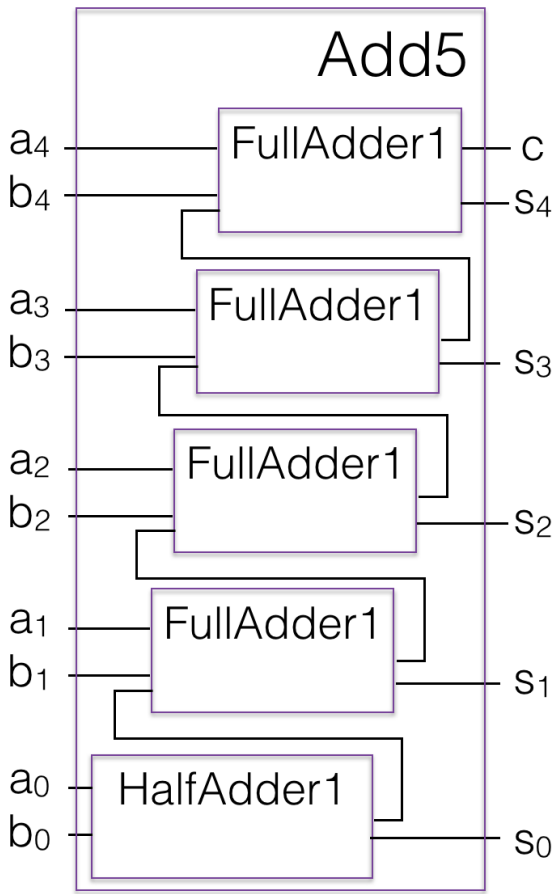
What's the simplest gate that matches this truth table? OR!

Full full adder



5-bit-number adder

Now, we can do the fundamental operation for adding multi-bit values. Can daisy chain them to add multi-bit numbers.



Deep circuits

Note that are circuits are getting kind of deep now.
How deep?

- Add5 is 1 HalfAdder1 plus 4 FullAdder1 deep.
- FullAdder1 is 2 HalfAdder1 plus 1 OR deep.
- HalfAdder1 is 1 XOR2 deep (and 1 AND in parallel).
- XOR2 is 1 IfThenElse1 plus 2 NOT deep.
- IfThenElse1 is 1 NOT, 1 AND, and 1 OR deep.

That's 49 gates deep, if you work it out.

Time to compute grows with the depth of the circuit. Fortunately, there are better designs for adding.

Negative numbers

We've defined addition in terms of logic gates. How about subtraction? We can compute subtraction via adding a negative number. What's a negative number in bits...?

$-x$ is a value such that $x + (-x) = 0$.

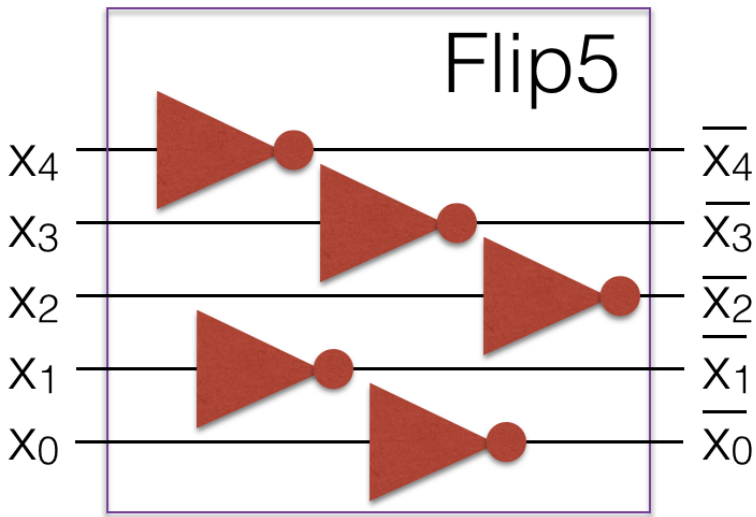
In computers, when we add two 5-bit numbers, we generally store a 5-bit answer, which means throwing away the carry. Thus, if we want the sum of two 5-bit numbers to result in a zero, we get that if they add to 32. So, to negate x, we can compute $32 - x$ (or just 0 if $x = 0$).

Example: $-5 \rightarrow 27$.

In binary: -00101 $\rightarrow$ 11011.
Progress? Well, we have a way to represent a negative number, but computing it seems to require subtraction, which is what we were trying to figure out in the first place...

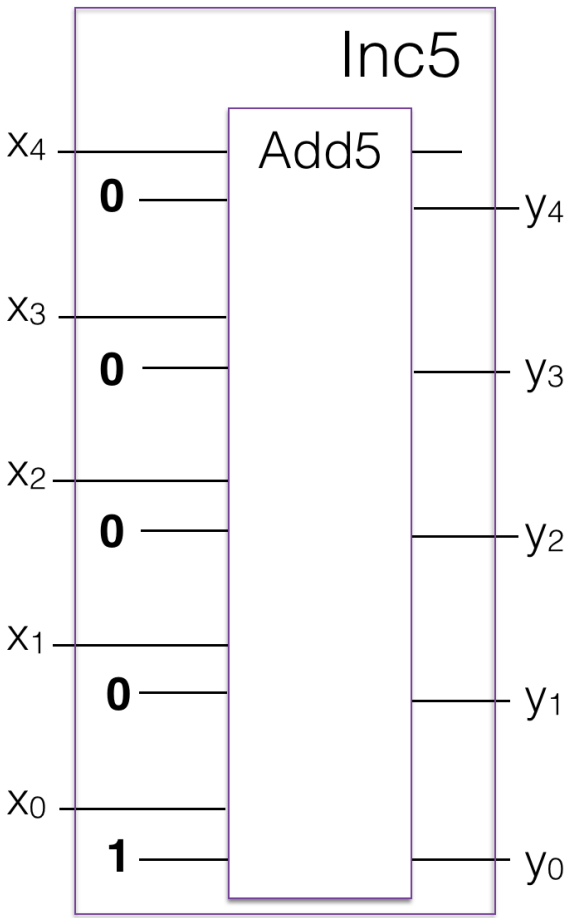
Bit flips

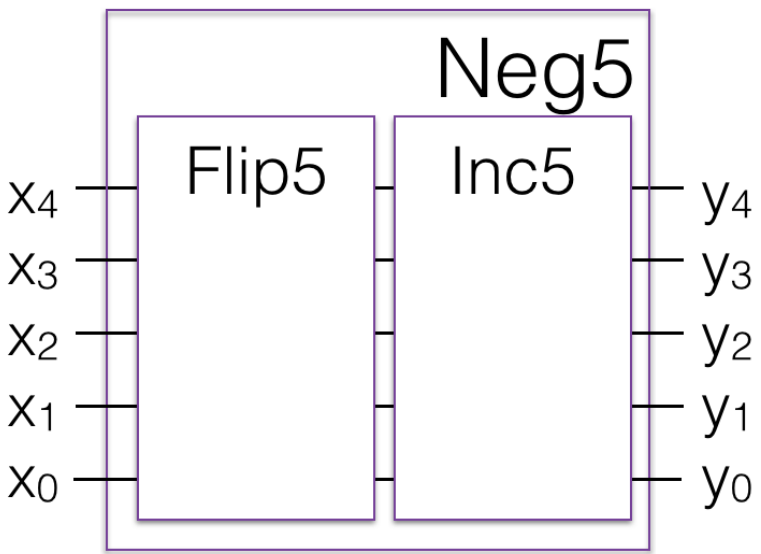
Define $\overline{x}$ to be the number x but with all its bits flipped.



Note: $x + \overline{x} = 2^5 - 1 = 31$ since the sum has every bit turned on. Thus, $\overline{x} = 31 - x$. If we add one to both sides, we have $\overline{x} + 1 = 32 - x$. And, $32 - x$ is the additive inverse, which is what we wanted to compute.

Negation by two's complement



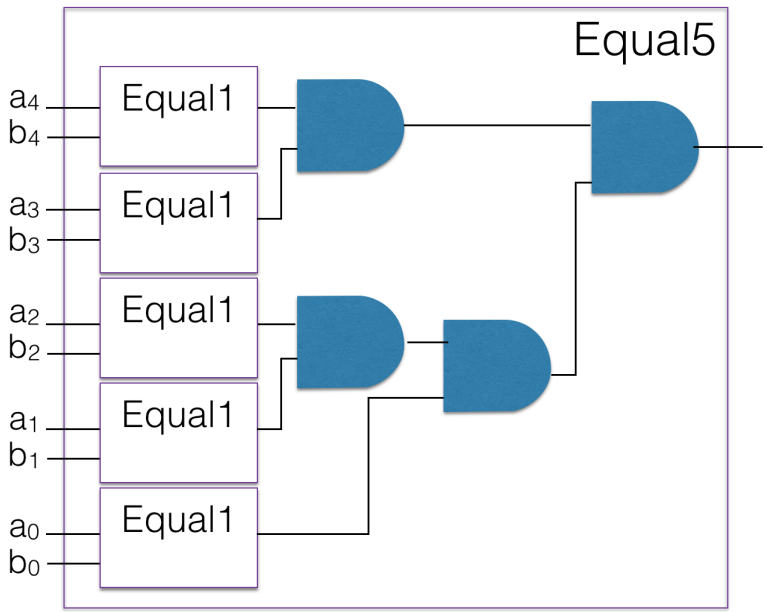


Positive and Negative Numbers

0	00000			
1	00001	11111	-1	(was 31)
2	00010	11110	-2	(was 30)
3	00011	11101	-3	(was 29)
4	00100	11100	-4	(was 28)
	...			
13	01101	10011	-13	(was 19)
14	01110	10010	-14	(was 18)
15	01111	10001	-15	(was 17)
		10000	-16	(was 16)

We interpret the numbers this way so we can use the first bit (high order bit) as the sign (0 for positive, 1 for negative).

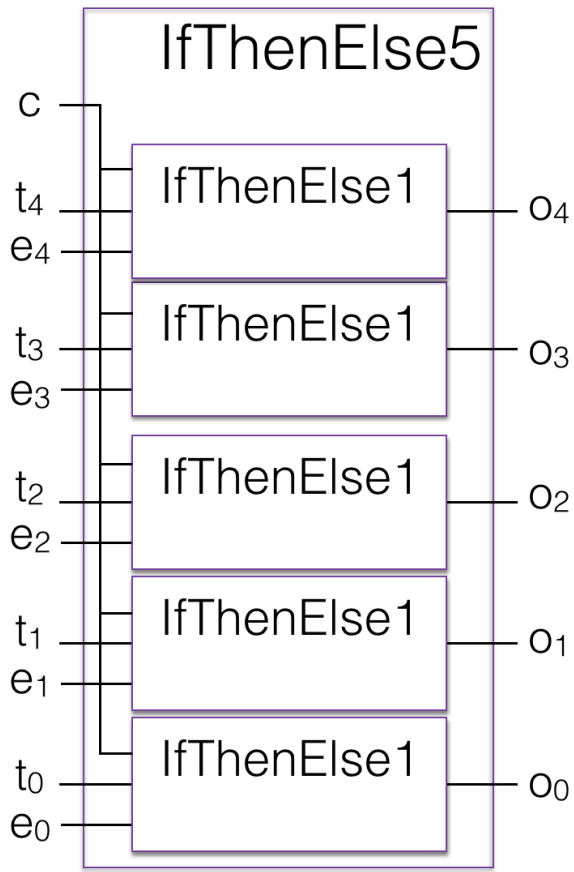
Equal5



Takes two 5-bit values $a_4 a_3 a_2 a_1 a_0$ and $b_4 b_3 b_2 b_1 b_0$. Returns **T** if and only if $a_i = b_i$ for all i.

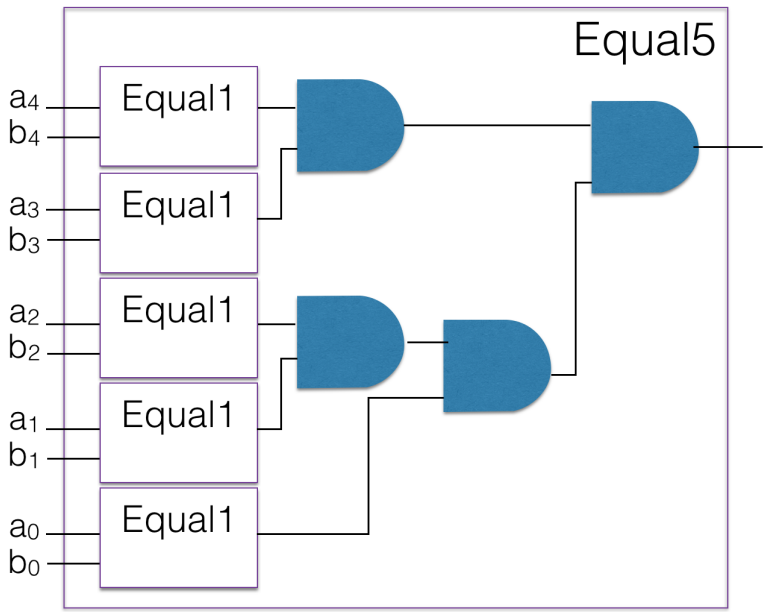
IfThenElse5

Now that we can add and subtract 5-bit numbers, we might want to perform other operations on 5-bit numbers as well.



Depending on condition bit c, produces either 5-bit t or 5-bit e.

Equal5



Takes two 5-bit values $a_4 a_3 a_2 a_1 a_0$ and $b_4 b_3 b_2 b_1 b_0$. Returns **T** if and only if $a_i = b_i$ for all i.